

CPSC 304 Project Cover Page

Milestone #: 4

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Group Number: 21

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Description:

Our project is a recipe management system. Users can upload, edit, and view various recipes. Recipes have ingredients, equipment, steps, and different tags (like dietary restrictions and cuisine types). Users can also sign up for courses taught by professional chefs who demo a recipe. We implemented features for users to filter recipes, filter courses, and view course registration statistics.

Final Schema Changes:

- Changed TEXT to varchar(500) for the description of recipe, as this was easier to work with in JavaScript
- Some small changes in the naming for consistency purposes
- No other changes

Copy of Schema:

Primary keys underlined, foreign keys bolded

recipe(rID: int, **uID**: int, title: varchar(50), description: varchar(500), servings: int) uID is not null

user(uID: int, email: varchar(50), uName: varchar(50)) email is unique and a candidate key

favourites(**uID**: int, **rID**: int)

professionalOne(YOE: int, rank: varchar(10))

professionalTwo(**uID**: int, YOE: int, speciality: varchar(30))

ingredient(iName: varchar(30), type: varchar(30))

hasTag(**tName**: varchar(30), **rID**: int)

equipment(eName: varchar(50), whereToBuy: varchar(50))

registers(**uID**: int, **cID**: int, date: date)

demosRecipe(**rID**: int, **cID**: int)

containsOne(**iName**: varchar(30), amount: varchar(30), cost: number(5, 2))

containsTwo(**rID**: int, **iName**: varchar(30), amount: varchar(30))

tag(tName: varchar(50), type: varchar(50))

step(**rID**: int, stepNumber: int, image: varchar(50), description: text)

requiresEq(**rID**: int, **eName**: varchar(50))

courseOne(**teacherID**: int, duration: int, difficulty: varchar(10), price: number(5, 2)) teacherID is not null

courseTwo(cID: int, **teacherID**: int, cName: varchar(30), difficulty: varchar(10), duration: int) teacherID is not null

review(reviewID: int, **rID**: int, **uID**: int, rating: int, comment: text) rID and uID are not null, (rID, uID) is unique and a candidate key

Data Present After SQL Script Runs:

USERS:

	USERID	EMAIL	UNAME
1	101	johnsmith@ubc.ca	John Smith
2	102	maryjane@gmail.com	Mary Jane
3	103	jpark@yahoo.ca	Jerry Park
4	104	bobbuilder@ubc.ca	Bob Builder
5	105	ssunnypatch@sfu.ca	Sarah Sunnypatch
6	106	coolgirl@sfu.ca	Cool Person

TAG:

	TNAME	TYPE
1	Italian	region
2	Pasta	dish
3	Vegetarian	diet
4	Dessert	course
5	Quick	time

STEP:

	RID	STEPNUMBER	IMAGE	DESCRIPTION
1	102	1	<null>	prepare the cheese
2	102	2	<null>	prepare the mac
3	102	3	<null>	mix them
4	101	1	<null>	melt the chocolate
5	101	2	<null>	put in in the oven

REVIEW:

	REVIEWID	RID	USERID	RATING	COMMENTBODY
1		101	101	102	4 It was too sweet
2		102	102	103	8 Easy and delicious
3		103	103	103	4 It would be better with more salt
4		104	101	104	9 Reminded me of the cookies my gra...
5		105	104	104	10 I totally loved it

REQUIRESEQ:

	RID	ENAME
1	101	handmixer
2	101	ladle
3	101	spoon
4	101	whisk
5	104	cast iron skillet
6	104	mandolin
7	105	whisk

REGISTERS:

	USERID	CID	REGISTRYDATE
1	101	101	2025-08-01
2	102	101	2025-08-01
3	101	102	2025-08-02
4	103	101	2025-08-03
5	101	103	2025-08-03
6	102	102	2025-08-03
7	101	105	2025-08-03
8	101	104	2025-08-03
9	102	104	2025-08-03

RECIPE:

	RID ▾	:	TITLE ▾	:	DESCRIPTION ▾	:	USERID ▾	:	SERVINGS ▾	:
1			101	Chocolate Chip Cookies	Moist and soft chocolate chip cookies.		101		10	
2			102	Mac and Cheese	Cheesey mac and cheese everyone will love.		102		4	
3			103	Veggie Stir-fry	Quick and healthy stir-fried vegetables.		102		2	
4			104	Spaghetti Bolognese	Classic Italian pasta with rich meat sauce		103		2	
5			105	Chocolate Cake	Decadent chocolate cake		103		10	

PROFESSIONAL_TWO:

	USERID ▾	⇅	YOE ▾	⇅	SPECIALITY ▾	⇅
1			101		5	Italian cuisine
2			102		10	Vegetarian recipes
3			103		1	Greek cuisine
4			104		2	Baker
5			105		6	Asian cuisine

PROFESSIONAL_ONE:

	YOE ▾	⇅	RANK ▾	⇅
1			1	Junior
2			2	Junior
3			5	Mid
4			6	Mid
5			10	Senior

INGREDIENT:

	INAME ▾	⇅	TYPE ▾	⇅
1	corn		vegetable	
2	macaroni		carbohydrate	
3	chicken		protein	
4	flour		baking	
5	pepper		seasoning	

HASTAG:

	RID ▾	⇅	TNAME ▾	⇅
1			101	Dessert
2			103	Quick
3			103	Vegetarian
4			104	Italian
5			104	Pasta

FAVOURITES:

	 RID 		 USERID 	
1		101		102
2		102		101
3		103		101
4		104		102
5		105		101

EQUIPMENT:

	 ENAME 		 WHERETOBUY 	
1	whisk		Ikea	
2	handmixer		Home Depot	
3	cast iron skillet		Ikea	
4	mandolin		Canadian Tire	
5	ladle		Ikea	
6	spoon		Ikea	

DEMOSRECIPE:

	 RID 		 CID 	
1		101		101
2		101		102
3		101		104
4		105		103
5		105		104

COURSE TWO:

	 CID 		 CNAME 		 TEACHERID 		 DURATION 		 DIFFICULTY 	
1		101	Desserts		101		10		easy	
2		102	Meals for Students		103		10		easy	
3		103	Everything Veggie		102		5		medium	
4		104	A Nice Meal for a Nice Date		105		15		medium	
5		105	Cook It and Lean It		102		20		hard	

COURSE ONE:

	TEACHERID	DURATION	DIFFICULTY	PRICE
1	101	10	easy	9.99
2	102	20	hard	99.99
3	103	10	easy	4.99
4	102	5	medium	44.99
5	105	15	medium	19.99

CONTAINSTWO:

	INAME	RID	AMOUNT
1	flour	101	50g
2	flour	105	100g
3	corn	103	1 cup
4	pepper	104	1 tsp
5	macaroni	102	1.5 cups

CONTAINSONE:

	INAME	AMOUNT	COST
1	flour	30g	2.11
2	corn	1 cup	4.50
3	macaroni	2 cups	5.00
4	chicken	3 drumsticks	9.50
5	pepper	0.5 tsp	1.00

Queries:

Insert: Inserting tag, [appService.js](#), line 472 - 492

Delete: Deleting recipe, [appService.js](#), line 697 - 709, init.sql line 60

Update: Updating recipe attributes, [appService.js](#), line 574 - 624

Selection: Filtering recipes, [appService.js](#), line 87 - 98

Join: Searching equipment store, [appService.js](#), line 167 - 177

Aggregation with group by: Counting course registration, [appService.js](#), line 115 - 127

Aggregation with having: Filtering courses by registration count, [appService.js](#), line 100 - 112

Division: Getting users in all courses, [appService.js](#), line 128 - 140

Nested aggregation with group by: Getting average registration by course difficulty, [appService.js](#), line 191 - 206

Projection: User chooses which recipe attributes to see, [appService.js](#), 219 - 230

Screenshots:

Insert:

Before:

```
SQL> SELECT * FROM HASTAG;
```

RID	TNAME
101	Dessert
103	Quick
103	Vegetarian
104	Italian
104	Meat
104	Pasta
105	cake

7 rows selected.

```
SQL> SELECT * FROM TAG;
```

TNAME	TYPE
Italian	region
Pasta	dish
Vegetarian	diet
TNAME	TYPE
Dessert	course
Quick	time

During:

Tag inserted!

After:


```
SQL> SELECT * FROM TAG;
```

TNAME	
TYPE	
Snack	snack type
cake	
ads	
Meat	poultry

```
SQL> SELECT * FROM HASTAG;
```

RID	TNAME
101	Dessert
101	Snack
103	Quick
103	Vegetarian
104	Italian
104	Meat
104	Pasta
105	cake

8 rows selected.

```
SQL> SELECT * FROM TAG;
```

TNAME	
TYPE	
Italian	region
Pasta	dish
Vegetarian	diet

```
SQL> SELECT * FROM TAG;
```

TNAME	
TYPE	
Dessert	course
Quick	time

```
SQL> SELECT * FROM TAG;
```

8 rows selected.

Delete:

Before:

```
RID TITLE
-----
DESCRIPTION
-----
USERID  SERVINGS
-----
101 Chocolate Chip Cookies
Moist and soft chocolate chip cookies.
101      10
102 Mac and Cheese
Cheesey mac and cheese everyone will love.
102      4
RID TITLE
-----
DESCRIPTION
-----
USERID  SERVINGS
-----
103 Veggie Stir-fry
Quick and healthy stir-fried vegetables.
103      2
104 Spaghetti Bolognese
Classic Italian pasta with rich meat sauce
RID TITLE
-----
DESCRIPTION
-----
USERID  SERVINGS
-----
103      2
105 Chocolate Cake
Decadent chocolate cake.
103      10
```

```
SQL> SELECT * FROM STEP;
```

RID	STEPNUMBER	IMAGE
102	1	
prepare the cheese		
102	2	
prepare the mac		
102	3	
mix them		

```
SQL> SELECT * FROM STEP;
```

RID	STEPNUMBER	IMAGE
101	1	
melt the chocolate		
101	2	
put in in the oven		

During:

Delete Recipe

Recipe ID:

Submit

After:

RID	TITLE
DESCRIPTION	
USERID	SERVINGS
101	Chocolate Chip Cookies
Moist and soft chocolate chip cookies.	
101	10
103	Veggie Stir-fry
Quick and healthy stir-fried vegetables.	
102	2
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
104	Spaghetti Bolognese
Classic Italian pasta with rich meat sauce	
103	2
105	Chocolate Cake
Decadent chocolate cake	
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
103	10

```
SQL> SELECT * FROM STEP;
```

RID	STEPNUMBER	IMAGE
DESCRIPTION		
101	1	
melt the chocolate		
101	2	
put in in the oven		

Update:

Before:

```
SQL> SELECT * FROM RECIPE;
```

RID	TITLE
DESCRIPTION	
USERID	SERVINGS
101	Chocolate Chip Cookies
Moist and soft chocolate chip cookies.	
101	10
102	Mac and Cheese
Creamy Mac and cheese everyone will love.	
102	4
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
103	Veggie Stir-fry :D
Quick and healthy stir-fried vegetables.	
102	2
104	Spaghetti Bolognese
Classic Italian pasta with rich meat sauce	
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
103	2
105	Chocolate Cake
Decadent chocolate cake	
103	10

During:

Change Recipe Title:

Old Title:

New Title:

Update Title

Title updated successfully!

Change Description:

Old Description:

New Description:

Update Description

Description updated successfully!

Change Number of Servings:

Old Servings:

New Servings:

Update Servings

Servings updated successfully!

Change Recipe Owner (User):

Old User:

New User:

Update User ID

User updated successfully!

After:

```
SQL> select*from recipe;

  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    101 Brownies
Moist and soft brownies!
    102         15

    102 Mac and Cheese
Cheesey mac and cheese everyone will love.
    102         4

  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    103 Veggie Stir-fry :D
Quick and healthy stir-fried vegetables.
    102         2

    104 Spaghetti Bolognese
Classic Italian pasta with rich meat sauce

  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    103         2

    105 Chocolate Cake
Decadent chocolate cake
    103        10
```

Selection:

Before:

```
  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    101 Chocolate Chip Cookies
Moist and soft chocolate chip cookies.
    101         10

    102 Mac and Cheese
Cheesey mac and cheese everyone will love.
    102         4

  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    103 Veggie Stir-fry
Quick and healthy stir-fried vegetables.
    102         2

    104 Spaghetti Bolognese
Classic Italian pasta with rich meat sauce

  RID TITLE
-----
DESCRIPTION
-----
  USERID  SERVINGS
-----
    103         2

    105 Chocolate Cake
Decadent chocolate cake
    103        10
```

During:

Recipes

Title contains:Chocolate

And ▾

Serving more than:1

And ▾

Serving less than:100

Filter

Chocolate Chip Cookies

Delete Recipe

Chocolate Cake

After:

```
SQL> SELECT * FROM RECIPE WHERE title LIKE '%Chocolate%' and servings > 1 and servings < 100;

RID TITLE
-----
DESCRIPTION
-----
USERID  SERVINGS
-----
101 Chocolate Chip Cookies
Moist and soft chocolate chip cookies.
101      10
105 Chocolate Cake
Decadent chocolate cake
103      10

RID TITLE
-----
DESCRIPTION
-----
USERID  SERVINGS
-----
```

Join:

Before:

```
SQL> SELECT * FROM REQUIRESEQ;

RID ENAME
-----
101 handmixer
101 ladle
101 spoon
101 whisk
104 cast iron skillet
104 mandolin
105 whisk
```

```
SQL> SELECT * FROM EQUIPMENT;

ENAME
-----
WHERETOBUY
-----
whisk
Ikea

handmixer
Home Depot

cast iron skillet
Ikea

ENAME
-----
WHERETOBUY
-----
mandolin
Canadian Tire

ladle
Ikea

spoon
Ikea

6 rows selected.
```

During:

Equipment

Item	Buy from
whisk	Ikea
ladle	Ikea
spoon	Ikea

After:

ENAME

WHERETOBUY

whisk
Ikea
ladle
Ikea
spoon
Ikea

Aggregation with group by:

Before:

```
SQL> SELECT * FROM REGISTERS;
```

USERID	CID	REGISTRYD
101	101	01-AUG-25
102	101	01-AUG-25
101	102	02-AUG-25
103	101	03-AUG-25
101	103	03-AUG-25
102	102	03-AUG-25
101	105	03-AUG-25
101	104	03-AUG-25
102	104	03-AUG-25

9 rows selected

During:

Count Registrations for each Course

Course: Students registered:

105	1
104	2
101	3
103	1
102	2

After:

```
SQL> SELECT CID, COUNT(*) FROM REGISTERS GROUP BY CID;
```

CID	COUNT(*)
105	1
104	2
101	3
103	1
102	2

Aggregation with having:

Before:

```
SQL> SELECT * FROM REGISTERS;
```

USERID	CID	REGISTRYD
101	101	01-AUG-25
102	101	01-AUG-25
101	102	02-AUG-25
103	101	03-AUG-25
101	103	03-AUG-25
102	102	03-AUG-25
101	105	03-AUG-25
101	104	03-AUG-25
102	104	03-AUG-25

9 rows selected

During:

Show courses with registration more than:

[Show](#)

After:

```
SQL> SELECT cID FROM REGISTERS GROUP BY cID HAVING COUNT(*) > 2;
```

CID
101

Division:

Before:

```
SQL> SELECT * FROM REGISTERS;
```

USERID	CID	REGISTRYD
101	101	01-AUG-25
102	101	01-AUG-25
101	102	02-AUG-25
103	101	03-AUG-25
101	103	03-AUG-25
102	102	03-AUG-25
101	105	03-AUG-25
101	104	03-AUG-25
102	104	03-AUG-25

9 rows selected

During:

[View Users Registered For All Courses](#)

User ID

101

After:

```
SQL> SELECT userID FROM USERS u WHERE NOT EXISTS (SELECT c.cID FROM courseTwo c MINUS SELECT r.cID FROM registers r WHERE r.userID = u.userID);
```

USERID
101

Nested Aggregation with Group By:

Before:

```
SQL> SELECT*FROM REGISTERS;
```

USERID	CID	REGISTRYD
101	101	01-AUG-25
102	101	01-AUG-25
101	102	02-AUG-25
103	101	03-AUG-25
101	103	03-AUG-25

During:

Average Registration By Course Difficulty

Difficulty: Average Registration:

easy

2

medium

1

After:

```
SQL> SELECT DIFFICULTY, AVG(USER_COUNT) FROM (SELECT COURSETWO.CID, DIFFICULTY, COUNT(UID) AS USER_COUNT FROM REGISTERS, COURSETWO WHERE REGISTERS.CID=COURSETWO.CID GROUP BY COURSETWO.CID, DIFFICULTY) GROUP BY DIFFICULTY;
```

DIFFICULTY	AVG(USER_COUNT)
easy	2
medium	1

Projection:

Before:

RID	TITLE
DESCRIPTION	
USERID	SERVINGS
101	Chocolate Chip Cookies
Moist and soft chocolate chip cookies.	
101	10
102	Mac and Cheese
Cheesey mac and cheese everyone will love.	
102	4
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
103	Veggie Stir-fry
Quick and healthy stir-fried vegetables.	
102	2
104	Spaghetti Bolognese
Classic Italian pasta with rich meat sauce	
RID	TITLE
DESCRIPTION	
USERID	SERVINGS
103	2
105	Chocolate Cake
Decadent chocolate cake	
103	10

During:

View Recipe Details

☒ Recipe ID

☒ Title

☐ Description

☐ Servings

☒ Creator ID

Submit

rID: 101

title: Chocolate Chip Cookies

userID: 101

After:

```
SQL> SELECT rID, userID, title FROM RECIPE;
```

RID	USERID	TITLE
101	101	Chocolate Chip Cookies
102	102	Mac and Cheese
103	102	Veggie Stir-fry
104	103	Spaghetti Bolognese
105	103	Chocolate Cake